

ABBOTSLEIGH
TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

MATHEMATICS

3 UNIT

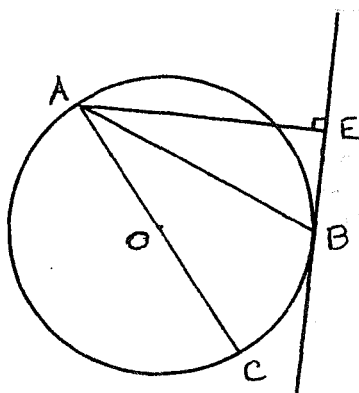
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Time allowed: Two hours
(Plus 5 minutes reading time)

Directions to candidates:

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied.
- Board-approved calculators may be used.
- Answer each question in a SEPARATE Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

- Q1. (a) Let $A(-5,12)$ and $B(4,9)$ be two points in the number plane. Find the coordinates of P which divides the interval AB externally in the ratio $5 : 2$. 2
- (b) Find the size of the acute angle between the lines $y = 2x + 3$ and $y = 4x + 1$. (Answer to the nearest minute). 2
- (c) Express $f(x) = x^3 + 3x^2 - 10x - 24$ as a product of three linear factors. 3
- (d) Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ 3
- (e) Two points A and B are placed on a circle and AC is a diameter. AE is perpendicular to the tangent at B . 2



- (i) Draw the diagram on your paper.
- (ii) Prove AB bisects $\angle CAE$.

Q2. Start a new booklet

- (a) Solve for x : $x \geq \frac{4}{x}$ 3
- (b) For $y = -3\sin^{-1} \frac{x}{2}$
- (i) State the domain and range. 3
- (ii) Sketch the curve.
- (c) Using the substitution $u = 9 - x^2$, evaluate $\int_0^3 x\sqrt{9-x^2} dx$ 3
- (d) The area bounded by the curve $y = \sin x$ between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x -axis. Find the volume of the solid of revolution. 3

Q3. Start a new booklet

- (a) Express $3\cos x + 4\sin x$ in the form $A\cos(x-\alpha)$ where $A > 0$. Hence, or otherwise, solve $3\cos x + 4\sin x = -3$ for $0 \leq x \leq 360^\circ$. 4
- (b) Find the greatest coefficient in the expansion $(3 + 4x)^{16}$ (leave in index form) 4
- (c) A point P moves on the curve $y = x^3$ in such a way that its x coordinate is changing at a constant rate of 2 units/sec. When $x = 1$, at what rate is
- (i) the y coordinate changing?
 - (ii) the gradient changing? 4

Q4. Start a new booklet

- (a) Find x and y if $\frac{4^x}{16} = 8^{x+y}$ and $2^{2x+y} = 128$. 3
- (b) If $x = 2 - \cos t$ and $y = 2t + 2\sin t$, 4
- (i) find $\frac{dx}{dt}$ and $\frac{dy}{dt}$
 - (ii) Hence or otherwise, find $\frac{dy}{dx}$ in terms of $\frac{t}{2}$.
- (c) A particle is oscillating in simple harmonic motion such that its displacement x metres from the origin is given by the equation $\frac{d^2x}{dt^2} = -9x$ where t is time in seconds. 5
- (i) Show that $x = a \cos(3t + \alpha)$ is a solution of motion for this particle (a and α are constants).
 - (ii) When $t = 0$, $v = 3$ m/s and $x = 5$ m. Show that the amplitude of the oscillation is $\sqrt{26}$ metres.
 - (iii) What is the maximum speed of the particle?

Q5. Start a new booklet

- (a) α, β, γ are the roots of the equation $2x^3 + 3x^2 - 4 = 0$ 3

Find

- (i) $\alpha + \beta + \gamma$
- (ii) $\alpha \beta \gamma$
- (iii) $\alpha^2 + \beta^2 + \gamma^2$

- (b) For the function $y = x^2 - 2x + 1$, find the largest possible domain such that this function has an inverse. Find the equation of this inverse and state its range. 3

- (c) For the parabola $x^2 = 12y$, find 6

- (i) the equation of the tangent at the point $P(6p, 3p^2)$ on the parabola.
- (ii) the coordinates of the point T where the tangent meets the x axis.
- (iii) Show that N , the midpoint of PT , has coordinates $(\frac{9p}{2}, \frac{3p^2}{2})$.
- (iv) Find the equation of the locus of N .

Q6. Start a new booklet

- (a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 2

- (b) The daily growth of a colony of insects is 10% of the excess of the population over 1.2×10^6 . 4
ie $\frac{dN}{dt} = 0.1 (N - 1.2 \times 10^6)$.

Initially, the population is 2.7×10^6 ,

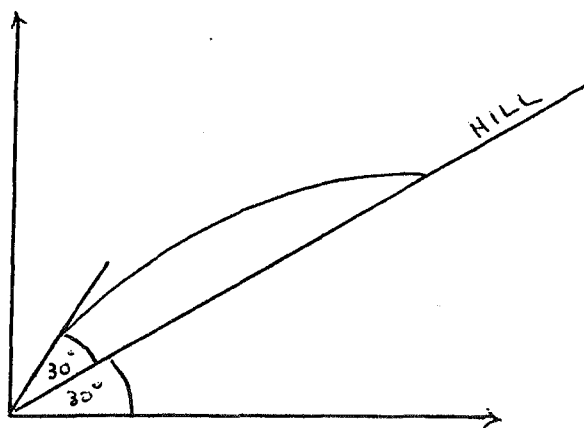
- (i) Determine the population after $3\frac{1}{2}$ days.
- (ii) If a scientist checks the population each day, which is the first day on which she should notice that the original population has tripled?

Q6. (continued).....

- (c) A ball is thrown with a velocity of $30\sqrt{3}$ m/s at an angle of 60° to the horizontal.

6

- (i) Assuming negligible air resistance and letting $g = 10 \text{ ms}^{-2}$, derive the equations of motion.
- (ii) Find the time of flight and the range.
- (iii) If the ball had been thrown with velocity $30\sqrt{3}$ m/s at an angle of 30° to a hill which is itself inclined at 30° to the horizontal (see diagram), determine the time of flight.



Q 7. Start a new booklet

- (a) Prove by mathematical induction that for all values of n

5

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!}$$

where n is a positive integer.

- (b) (i) Show that $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ has no stationary points.

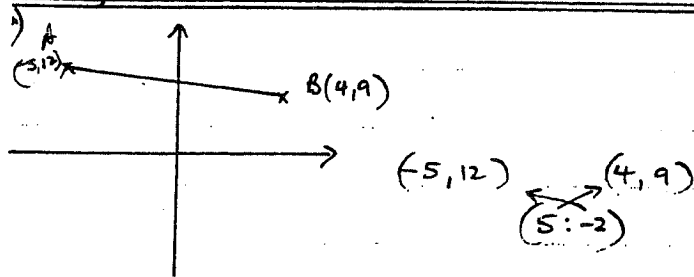
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(ii) Prove that the lines $y = \pm 1$ are asymptotes.

(iii) Sketch the curve.

(iv) If k is a positive constant, find the area in the first quadrant enclosed by the above curve and the three lines $y = 1$, $x = 0$ and $x = k$.

(v) Prove that for all values of k , this area is always less than $\log_e 2$.



$$P = \left(\frac{-2x - 5 + 5 \times 4}{5 - 2}, \frac{-2 \times 12 + 5 \times 9}{5 - 2} \right)$$

$$P = (10, 7)$$

b) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ $m_1 = 2$ $m_2 = 4$

$$= \frac{4 - 2}{1 + 2 \times 4}$$

$$= \frac{2}{9}$$

$$\theta = 12^\circ 32'$$

c) $f(x) = x^3 + 3x^2 - 10x - 24$

$$f(-2) = -8 + 12 + 20 - 24 = 0$$

$$\begin{array}{r} x^2 + x - 12 \\ x+2 \overline{) x^3 + 3x^2 - 10x - 24} \\ \underline{x^3 + 2x^2} \\ x^2 - 10x \\ \underline{x^2 + 2x} \\ -12x - 24 \\ \underline{-12x - 24} \\ 0 \end{array}$$

$$f(x) = (x+2)(x^2 + x - 12)$$

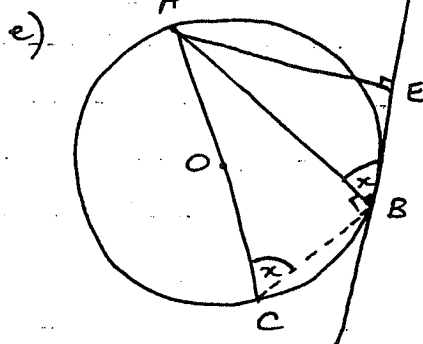
$$= (x+2)(x+4)(x-3)$$

d) $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

$$= \left[\sin^{-1} \frac{x}{3} \right]_0^3$$

$$= \sin^{-1} 1 - \sin^{-1} 0$$

$$= \frac{\pi}{2}$$



$\angle ABE = \angle ACB = x$ (angle in alternate segments equal)

$\angle ABC = 90$ (angle in semicircle = 90)

$\therefore \angle CAB = 90 - x$ (angle sum Δ)

$\angle BAE = 90 - x$ (angle sum Δ)

$\therefore \angle CAB = \angle BAE$ (both = $90 - x$)

$\therefore AB$ bisects $\angle CAE$

2a) $x^2 \times x \geq \frac{4}{x} \times x^2$ $x \neq 0$

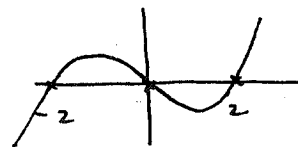
$$x^3 \geq 4x$$

$$x^3 - 4x \geq 0$$

$$x(x^2 - 4) \geq 0$$

$$x(x-2)(x+2) \geq 0$$

$$\therefore x \geq 2 \quad -2 \leq x < 0$$



b) $y = -3 \sin^{-1} \frac{x}{2}$

D: $-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

R: $-\pi \leq \sin^{-1} \frac{x}{2} \leq \pi$

$$c) \quad y = x^3 \quad \frac{dx}{dt} = 2 \text{ u/s}$$

$$\frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$= 3x^2 \times 2$$

$$\frac{dy}{dt} = 6x^2, \text{ when } x = 1$$

$$\frac{dy}{dt} = 6 \text{ u/s}$$

Let $m = \text{gradient}$

$$m = 3x^2$$

$$\frac{dm}{dx} = 6x$$

$$\frac{dm}{dt} = \frac{dm}{dx} \cdot \frac{dx}{dt}$$

$$= 6x \cdot 2$$

$$= 12x$$

$$\text{when } x = 1 \quad \frac{dm}{dt} = 12 \text{ /s}$$

rate of change of gradient is 12 per second

$$4a) \quad \frac{4^x}{16} = 8^{x+y} \quad 2^{2x+y} = 128$$

$$2^{2x-4} = 2^{3x+3y}$$

$$2^{2x+y} = 2^7$$

$$\therefore 2x+y = 7$$

$$\therefore 2x-4 = 3x+3y$$

$$-4 = x+3y$$

Solve. $x+3y = -4$

$$2x+y = 7 \quad \therefore y = 7-2x$$

$$x+3(7-2x) = -4$$

$$x+21-6x = -4$$

$$-5x = -25$$

$$x = 5$$

$$y = 7-10 = -3$$

$$b) \quad x = 2 - \cos t$$

$$y = 2t + 2\sin t$$

$$\frac{dx}{dt} = +\sin t$$

$$\frac{dy}{dt} = 2 + 2\cos t$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{2 + 2\cos t}{\sin t}$$

$$= \frac{2(1 + \cos t)}{\sin t}$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$\cos \theta = 2\cos^2 \frac{\theta}{2} - 1$$

$$= \frac{2(1 + (2\cos^2 \frac{t}{2} - 1))}{2\sin \frac{t}{2} \cos \frac{t}{2}}$$

$$= \frac{2\cos^2 \frac{t}{2}}{\sin \frac{t}{2} \cos \frac{t}{2}}$$

$$= \frac{2\cos \frac{t}{2}}{\sin \frac{t}{2}}$$

$$= 2 \cot \frac{t}{2}$$

$$\frac{dy}{dx} = \frac{2(1 + \cos t)}{\sin t} \quad \text{where } t = t$$

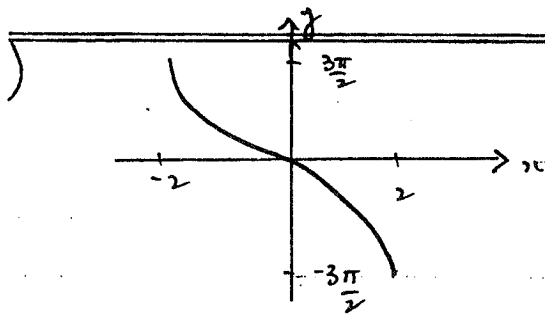
$$= 2 \left(1 + \frac{1-t^2}{1+t^2} \right) \div \frac{2t}{1+t^2}$$

$$= 2 \left(\frac{1+t^2+1-t^2}{1+t^2} \right) \times \frac{1+t^2}{2t}$$

$$= \frac{2}{t}$$

$$= \frac{2}{\tan \frac{t}{2}}$$

$$= 2 \cot \frac{t}{2}$$



$$1) \quad u = 9 - x^2 \quad \text{when } x=0 \quad u=9 \\ x=3 \quad u=0$$

$$\frac{du}{dx} = -2x$$

$$\int_0^3 x \sqrt{9-x^2} dx = \int_9^0 u^{\frac{1}{2}} x - \frac{1}{2} du$$

$$= \frac{1}{2} \int_0^9 u^{\frac{1}{2}} du$$

$$= \frac{1}{2} \cdot \frac{2}{3} \left[u^{\frac{3}{2}} \right]_0^9$$

$$= \frac{1}{3} [27 - 0]$$

$$= 9$$

$$2) \quad y = \sin x \\ V = \pi \int_0^{\pi/2} \sin^2 x dx \\ = \frac{\pi}{2} \int_0^{\pi/2} 1 - \cos 2x dx \\ = \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} \left[\frac{\pi}{2} - \frac{1}{2} \sin \pi - \left(0 - \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi}{2} \left[\frac{\pi}{2} \right]$$

$$= \frac{\pi^2}{4}$$

$$3a) \quad 3 \cos x + 4 \sin x$$

$$= A \cos x \cos \alpha + A \sin x \sin \alpha = A \cos(x - \alpha)$$

$$\therefore A \cos \alpha = 3$$

$$A \sin \alpha = 4$$

$$\tan \alpha > 0$$

$$\tan \alpha = \frac{4}{3}$$

$$\therefore \alpha \text{ is acute}$$

$$\alpha = 53^\circ 8'$$

$$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 3^2 + 4^2$$

$$A^2 = 25$$

$$A = 5$$

$$\therefore 3 \cos x + 4 \sin x = 5 \cos(x - 53^\circ 8')$$

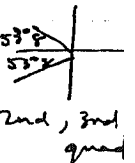
$$-53^\circ 8' \leq x - 53^\circ 8' \leq 36^\circ$$

$$3 \cos x + 4 \sin x = 5 \cos(x - 53^\circ 8') = -3$$

$$\cos(x - 53^\circ 8') = -\frac{3}{5}$$

$$x - 53^\circ 8' = 126^\circ 52', 233^\circ 8'$$

$$x = 180^\circ, 286^\circ 16'$$



2nd, 3rd quad

$$b) \quad (3+4x)^{16} \\ T_{k+1} = {}^{16}C_k 3^{16-k} 4^k \\ T_k = {}^{16}C_{k-1} 3^{17-k} 4^{k-1}$$

$$\frac{T_{k+1}}{T_k} \geq 1 \quad \text{for greatest coeff.}$$

$$\frac{{}^{16}C_k 3^{16-k} 4^k}{{}^{16}C_{k-1} 3^{17-k} 4^{k-1}} \geq 1$$

$$\frac{16! 3^{16-k} 4^k}{(16-k)! k!} \times \frac{(17-k)!(k-1)!}{16! 3^{17-k} 4^{k-1}} \geq 1$$

$$\frac{(17-k) 4}{3k} \geq 1$$

$$68 - 4k \geq 3k$$

$$7k \leq 68$$

$$k \leq 9 \frac{5}{7}$$

$$\therefore k = 9$$

$$\therefore \frac{d^2x}{dt^2} = -9x$$

$$) \quad x = a \cos(3t + \alpha)$$

$$\frac{dx}{dt} = -3a \sin(3t + \alpha)$$

$$\frac{d^2x}{dt^2} = -9a \cos(3t + \alpha) = -9x$$

$\therefore x = a \cos(3t + \alpha)$ is a solⁿ

$$x = a \cos(3t + \alpha)$$

$$i) \quad v^2 = n^2(a^2 - x^2)$$

$$3^2 = 3^2(a^2 - 5^2)$$

$$1 = a^2 - 25$$

$$a^2 = 26$$

$$a = \sqrt{26}$$

$$\begin{aligned} \text{R} \quad x &= a \cos(3t + \alpha) & \dot{x} &= -3a \sin(3t + \alpha) \\ 5 &= a \cos \alpha \quad \text{when } t=0 & 3 &= -3a \sin \alpha \\ \cos \alpha &= \frac{5}{a} & \sin \alpha &= -\frac{1}{a} \end{aligned}$$

$$\cos^2 \alpha + \sin^2 \alpha = 1$$

$$\frac{25}{a^2} + \frac{1}{a^2} = 1$$

$$a^2 = 26 \quad \therefore a = \sqrt{26}$$

) max speed when

$$\sin(3t + \alpha) = 1$$

$$v = -3\sqrt{26} \sin(3t + \alpha) \text{ is a max}$$

$$\text{when } \sin(3t + \alpha) = 1$$

$$\text{i.e. } v = -3\sqrt{26}$$

$$\therefore \text{max speed is } 3\sqrt{26} \text{ m/s}$$

$$\text{R} \quad v^2 = n^2(a^2 - x^2) \text{ for SHM}$$

max velocity occurs when $x=0$

$$v^2 = 9(26 - 0)$$

$$v = \pm 3\sqrt{26}$$

$$\text{max speed is } 3\sqrt{26} \text{ m/s}$$

$$5. \quad 2x^3 + 3x^2 - 4 = 0$$

$$a=2$$

$$b=3$$

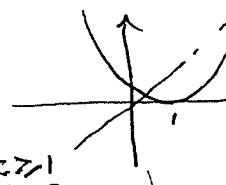
$$c=-4$$

$$d=-4$$

$$\alpha + \beta + \gamma = -\frac{3}{2}$$

$$\alpha\beta\gamma = \frac{4}{2} = 2$$

$$\begin{aligned} (\alpha + \beta + \gamma)^2 &= \alpha^2 + \beta^2 + \gamma^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(-\frac{3}{2}\right)^2 + 2 \times 0 \\ &= \frac{9}{4} \end{aligned}$$



$$b) \quad y = x^2 - 2x + 1$$

$$= (x-1)^2$$

$$D: x \geq 1$$

$$R: y \geq 0$$

D: $x \geq 0$
R: $y \geq 1$ for inverse f^{-1}

for inverse f^{-1}

$$x = (y-1)^2$$

$$\pm x^{\frac{1}{2}} = y-1$$

$$y = 1 \pm \sqrt{x}$$

But $y \geq 1$

$\therefore$ inverse f^{-1} is

$$y = 1 + \sqrt{x}$$

$$c) \quad x^2 = 12y$$

$$i) \quad y = \frac{x^2}{12}$$

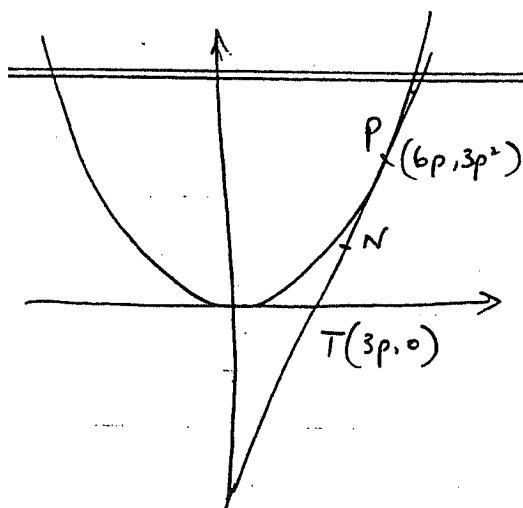
$$\frac{dy}{dx} = \frac{2x}{12} = \frac{x}{6}$$

$$\text{when } x = 6p \quad \frac{dy}{dx} = \frac{6p}{6} = p$$

$$\text{eqn tangent } \frac{y-3p^2}{x-6p} = p$$

$$y - 3p^2 = px - 6p^2$$

$$-y = px - 3p^2$$



$$y = px - 3p^2$$

o. and of T when $y = 0$ $y = px - 3p^2$

$$px - 3p^2 = 0$$

$$p(x - 3p) = 0 \quad p = 0$$

$$x = 3p$$

$$T(3p, 0)$$

o. and of N

$$P(6p, 3p^2) \quad T(3p, 0)$$

$$N\left(\frac{6p+3p}{2}, \frac{3p^2}{2}\right)$$

$$\left(\frac{9p}{2}, \frac{3p^2}{2}\right)$$

lous of N

$$x = \frac{9p}{2} \quad y = \frac{3p^2}{2}$$

$$\therefore p = \frac{2x}{9}$$

$$y = \frac{3}{2} \left(\frac{2x}{9}\right)^2$$

$$= \frac{3}{2} \left(\frac{4x^2}{81}\right)$$

$$= \frac{2x^2}{27}$$

$\therefore 27y = 2x^2$ is eqn of lous of N.

$$6a) \lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$$

$$= \frac{1}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times 3$$

$$= \frac{3}{5}$$

$$b) \frac{dN}{dt} = 0.1(N - 1.2 \times 10^6)$$

$$N = P + Ae^{kt}$$

$$t=0 \quad N = 2.7 \times 10^6 \quad P = 1.2 \times 10^6$$

$$2.7 \times 10^6 = 1.2 \times 10^6 + Ae^0$$

$$1.5 \times 10^6 = A$$

$$\therefore N = 1.2 \times 10^6 + 1.5 \times 10^6 e^{0.1t}$$

i) when $t = 3.5$

$$N = 1.2 \times 10^6 + 1.5 \times 10^6 \times e^{0.1 \times 3.5}$$

$$= 3.32 \times 10^6$$

ii) $3N = 8.1 \times 10^6$

$$8.1 \times 10^6 = 1.2 \times 10^6 + 1.5 \times 10^6 \times e^{0.1t}$$

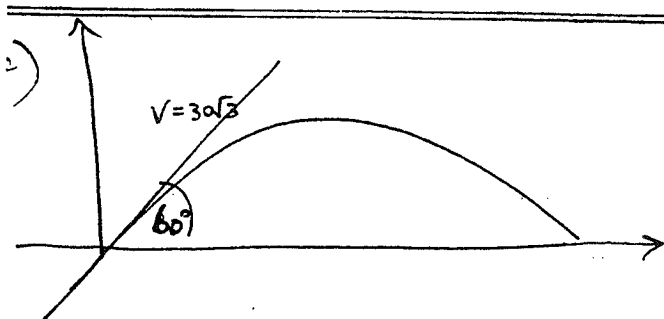
$$4.6 = e^{0.1t}$$

$$0.1t = \ln 4.6$$

$$t = \frac{\ln 4.6}{0.1}$$

$$= 15.2$$

$\therefore$ on 16th day the pop has tripled



$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

$$\text{when } t=0 \quad \dot{x} = 30\sqrt{3} \cos 60 \\ = 15\sqrt{3}$$

$$\therefore \dot{x} = 15\sqrt{3}$$

$$x = 15\sqrt{3}t + C_2$$

$$\text{when } t=0 \quad x=0 \quad \therefore C_2 = 0$$

$$x = 15\sqrt{3}t$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_3$$

$$\text{when } t=0 \quad \dot{y} = 30\sqrt{3} \sin 60 \\ = 45$$

$$\therefore \dot{y} = -10t + 45$$

$$y = -5t^2 + 45t + C_4$$

$$\text{when } t=0 \quad y=0 \quad \therefore C_4 = 0$$

$$\therefore y = -5t^2 + 45t$$

$$\text{for time of flight } y=0$$

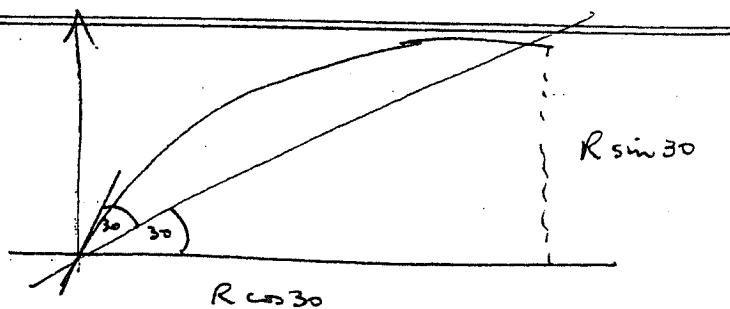
$$-5t^2 + 45t = 0$$

$$5t(-t + 9) = 0$$

$$t=0 \quad t=9 \quad \therefore \text{time of flight} = 9s$$

$$\text{for range } x = 15\sqrt{3}t \text{ at } t=9$$

$$x = 135\sqrt{3} \text{ m.}$$



$$x = R \cos 30 = \frac{R\sqrt{3}}{2} = 15\sqrt{3}t$$

$$y = R \sin 30 = \frac{R}{2} = -5t^2 + 45t$$

$$\therefore R = 30t$$

$$\frac{R}{2} = -5t^2 + 45t$$

$$15t = -5t^2 + 45t$$

$$5t^2 - 30t = 0$$

$$5t(t-6) = 0$$

$$t=0 \quad t=6$$

$\therefore$ time of flight is 6 secs.

OR

$$\tan 30 = \frac{R \sin \theta}{R \cos 30} = \frac{y}{x}$$

$$\frac{1}{\sqrt{3}} = \frac{-5t^2 + 45t}{15\sqrt{3}t}$$

$$15t = -5t^2 + 45t$$

$$5t^2 - 30t = 0$$

$$5t(t-6) = 0$$

$$\therefore t=0 \quad t=6$$

$\therefore$ time of flight is 6 secs.

$$2) \quad \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!} \quad \text{for all } n.$$

Step 1. Test for $n=1$

$$\text{LHS} = \frac{1}{2!}$$

$$= \frac{1}{2}$$

$$\text{RHS} = \frac{2! - 1}{2!}$$

$$= \frac{1}{2}$$

$\therefore$ true for $n=1$

Step 2

Assume true for $n=k$

$$\text{i.e.} \quad \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = \frac{(k+1)! - 1}{(k+1)!}$$

* show true for $n=k+1$

$$\text{i.e. show } \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = \frac{(k+2)! - 1}{(k+2)!}$$

$$\text{LHS} = \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)((k+1)! - 1) + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - (k+2) + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - k - 2 + k+1}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$= \text{RHS}$$

$\therefore$ if it is true for $n=k$, then it is true for $n=k+1$.

Step 3. Since it is true for $n=1$, by step 2, it must be true for $n=1+1=2$
 & since it is true for $n=2+1=3$ & so on $\therefore$ it is true for all n .

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = \frac{(n+1)! - 1}{(n+1)!} \quad \text{for all } n.$$

b) i) $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$y' = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + e^{-2x} + 2 - (e^{2x} + e^{-2x} - 2)}{(e^x + e^{-x})^2}$$

$$= \frac{4}{(e^x + e^{-x})^2}$$

$\neq 0$ for all x

$\therefore$ there are no stationary pts

ii) as $x \rightarrow \infty$ $e^{-x} \rightarrow 0$

$$\therefore y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \rightarrow \frac{e^x}{e^x} = 1$$

as $x \rightarrow -\infty$ $e^x \rightarrow 0$

$$\therefore y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \rightarrow \frac{-e^{-x}}{e^{-x}} = -1$$

$\therefore y = \pm 1$ are the asymptotes.

or

for $y = 1$

$$1 = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$e^x + e^{-x} = e^x - e^{-x}$$

$$2e^{-x} = 0 \quad \text{No sol}^n$$

$\therefore y = 1$ is an asymptote

for $y = -1$

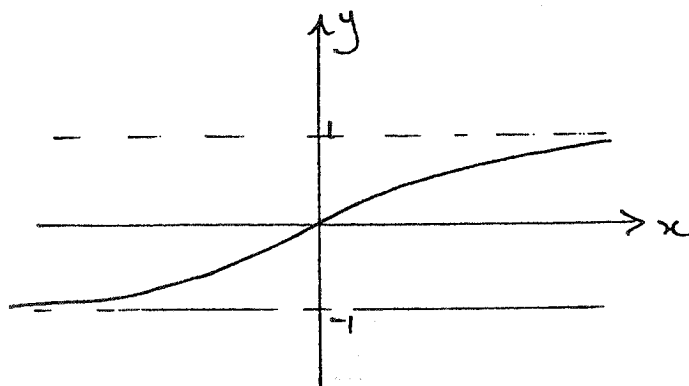
$$-1 = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$-e^x - e^{-x} = e^x - e^{-x}$$

$$2e^x = 0 \quad \text{No sol}^n$$

$\therefore y = -1$ is an asymptote

iii)



$$\text{iv) Area} = k - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

$$= k - \left[\log_e (e^x + e^{-x}) \right]_0^k$$

$$= k - \left[\log_e (e^k + e^{-k}) - \log_e (e^0 + e^0) \right]$$

$$= k - \left[\log_e (e^k + e^{-k}) - \log_e 2 \right]$$

$$\text{v) } k - \left[\log_e (e^k + e^{-k}) - \log_e 2 \right]$$

$$= \log_e e^k - \log_e (e^k + e^{-k}) + \log_e 2$$

$$= \log_e \frac{e^k}{e^k + e^{-k}} + \log_e 2$$

$$\text{Now } \log_e \frac{e^k}{e^k + e^{-k}} < 0 \quad \text{because } \frac{e^k}{e^k + e^{-k}} < 1$$

$\therefore$ max value is $\log_e 2$